



Mechanical Repairs Building TT113
at the
UNITED STATES MARINE CORPS BASE
CAMP LEJEUNE, NORTH CAROLINA

June 18, 2025

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BASIS OF DESIGN

Final Design Submittal



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REFERENCE DOCUMENTS

All work will comply with the latest edition of the following codes and standards:

ACI 318-19	Building Code Requirements for Structural Concrete
ACI 360R-10	Guide to Design of Slabs-on-Ground
ADA/ABA	Americans with Disabilities Act (ADA) and Architectural Barrier Act (ABA) Accessibility Guidelines
ASCE 7-16	Minimum Design Loads for Buildings and Other Structures
ASHRAE 15	Safety Standard for Refrigerated Systems, 2016 edition
ASHRAE 36	High-Performance Sequences of Operation for Ventilation Systems, 2021 edition
ASHRAE 55	Thermal Environmental Conditions for Human Occupancy, 2023 edition
ASHRAE 62.1	Ventilation for Acceptable Indoor Air Quality, 2022 edition
ASHRAE 90.1	Energy Standard for Sites and Buildings except Low Rise Residential Buildings, 2022 edition
ASTM	American Society for Testing and Materials
AWS D1.1-20	Structural Welding Code
IBC 2021	International Building Code
NCDENR	North Carolina Department of Environmental and Natural Resources
NFPA 54	National Fuel Gas Code, 2024 edition
NFPA 70	National Electric Code, 2023 edition
NFPA 90A	Standard for the Installation of Air-Conditioning and Ventilation System, 2024 edition
NFPA 101	Life Safety Code, 2024 edition
NFPA 780	Standard for the Installation of Lightning Protection Systems, 2023 edition
SMACNA	HVAC Duct Construction Standards - Metal and Flexible, 4 th Edition
SMACNA	HVAC Air Duct Leakage Test Manual, 2nd Edition
UFC 1-200-01	DoD Building Code, with Change 3, 26 February 2024
UFC 1-200-02	High Performance and Sustainable Building Requirements, with Change 2, 01 June 2022
FC 1-300-09N	Navy and Marine Corps Design Procedures, 17 May 2024
UFC 3-101-01	Architecture, with Change 4, 08 January 2024
UFC 3-110-03	Roofing, with Change 5, 12 June 2020
UFC 3-120-01	Design: Sign Standards, with Change 4, 01 February 2023
UFC 3-120-10	Interior Design, with Change 2, 15 June 2021
UFC 3-190-06	Protective Coatings and Paints, 01 March 2021
UFC 3-201-01	Civil Engineering, 20 December 2022
UFC 3-230-01	Water Storage, Distribution, and Transmission, with Change 3, 01 July 2021
UFC 3-240-01	Wastewater Collection and Treatment, with Change 4, 01 October 2024
UFC 3-250-01	Pavement Design for Roads and Parking Areas, 14 November 2016
UFC 3-250-04	Standard Practice for Concrete Pavement Construction, 29 January 2024
UFC 3-301-01	Structural Engineering, Change 2, 04 September 2024
UFC 3-400-02	Design: Engineering Weather Data, with Change 2, 20 September 2024
UFC 3-401-01	Mechanical Engineering, with Change 1, 01 October 2015

UFC 3-410-01 Heating, Ventilating, and Air Conditioning Systems, with Change 9, 09 January 2024
 UFC 3-420-01 Plumbing Systems, 01 April 2021
 UFC 3-450-01 Noise and Vibration Control, 08 December 2022
 UFC 3-501-01 Electrical Engineering, 27 November 2024
 UFC 3-520-01 Interior Electrical Systems, 16 January 2025
 UFC 3-530-01 Interior and Exterior Lighting Systems and Controls, Change 1, 15 December 2023
 UFC 3-550-01 Exterior Electrical Power Distribution, Change 3, 01 Nov 2019
 UFC 3-560-01 Operation and Maintenance: Electrical Safety, Change 3, 15 May 2023
 UFC 3-575-01 Lightning and Static Electricity Protection Systems, Change 1, 01 Oct 2021
 UFC 3-580-01 Telecommunications Building System Design, 21 November Jun 2024
 UFC 3-600-01 Fire Protection Engineering for Facilities, with Change 6, 06 May 2021
 UFC 4-010-01 DOD Minimum Antiterrorism Standards for Buildings, with Change 3, 24 May 2024
 UFC 4-410-02 DDC for HVAC, Change 2, 12 April 2021
 UFC 4-740-06 Youth Centers, with Change 1, 21 March 2022
 Camp Lejeune Mechanical Design Guidance – Aug 2022
 Camp Lejeune Electrical Policies_v20220308
 Applicable UL standards (latest editions)
 Applicable NEMA standards (latest editions)
 Camp Lejeune Base Specification 27 10 00, v04/22
 Camp Lejeune Base Specification 33 82 00, v08/22
 Applicable EIA and TIA standards (latest editions)
 Applicable UL standards (latest editions)

ARCHITECTURAL BASIS OF DESIGN

REFERENCE DOCUMENTS

Architectural work will comply with the latest edition of the following codes and standards:

International Building Code (IBC)

NFPA101 Life Safety Code (NFPA), 2023 Edition

UFC 1-200-01 DoD Building Code, with Change 3, 26 February 2024

FC 1-300-09N Design Procedures, 17 May 2024

UFC 3-101-01 Architecture, with Change 4, 08 January 2024

UFC 3-110-03 Roofing, with Change 5, 12 June 2020

UFC 3-120-10 Interior Design, with Change 2, 15 June 2021

UFC 3-190-06 Protective Coatings and Paints, 01 March 2021

UFC 3-450-01 Noise and Vibration Control, 08 December 2022

UFC 3-600-01 Fire Protection Engineering for Facilities, with Change 6, 06 May 2021

UFC 4-010-01 DOD Minimum Antiterrorism Standards for Buildings, with Change 3, 24 May 2024

Architectural Barriers Act Standards (ABA)

SCOPE OF WORK

This Basis of Design narrative outlines the repair and renovation of building TT113 and the construction of a new mechanical building and chiller yard adjacent to building TT113. Building TT113 is located in the Tarawa Terrace area of United States Marine Corps Base Camp Lejeune.

Building TT113 is a single-story, 31,664 square foot Childhood Development Center (CDC). The scope of renovation includes complete replacement of the existing HVAC system and selected interior, and exterior improvements as noted in the SAES. The renovations will improve occupant comfort of both students and employees of the CDC.

Exterior work includes the following:

- Construct a new mechanical building and yard.
- Remove existing standing seam metal roof, skylights, and all existing metal gutters, downspouts, fascia, and soffit. The roof will be infilled where skylights were removed, and a new standing seam metal roof, metal gutters, downspouts, fascia, and soffit will be installed.
- Metal panel siding and louvers on vertical dormer walls will be removed. New metal panel siding and louvers will be installed along with new metal roof flashing.
- A new roof dormers with a mechanical access door will be provided on the east, west, and south ends of the building for access to a new mechanical attic catwalk.
- Provide infrastructure for (2) new level 2 EV charging systems.

Interior work includes the following:

- Remove existing HVAC system in its entirety including all components, air handlers, ductwork, controls, etc. and replace with new VAV or 4-pipe fan coil HVAC system.
- Remove and replace all existing ceilings.

- Remove existing lightwell bulkheads and lightwell gypsum board and infill lightwell openings to provide separation between occupied space and attic. Infill ceilings will cover areas where lightwells are removed.
- The existing building backflow preventer will be relocated to accommodate and a new attic access ladder and hatch.
- A new mechanical access catwalk will be provided for circulation between attic mounted HVAC equipment.
- Remove and replace all existing lighting.
- Remove and replace all existing finished flooring.
- Paint all previously painted surfaced.
- Plumbing will be evaluated and modified as needed to correct current pressure and hot water supply issues.
- Evaluate kitchen HVAC and address current ventilation and humidity control conditions.

TYPE OF CONSTRUCTION

Type IIB non-combustible construction

OCCUPANCY CLASSIFICATIONS

NFPA 101 Occupancy: Day-care

LIFE SAFETY CODE ANALYSIS

Reference the Fire Protection Design Analysis

ACCESSIBILITY

The facility design meets the requirements of the Architectural Barriers Act and UFC 1-200-01.

ARCHITECTURAL COMPATIBILITY

Building renovations will comply with the MCB Camp Lejeune Base Exterior Architectural Plan (BEAP), dated 2010.

EXTERIOR WALL SYSTEM

The existing exterior wall construction consists of smooth-faced, modular-sized brick veneer backed up by concrete masonry unit walls. Construction assemblies integrate an air and moisture barrier at floors, walls and roofs.

ROOFING

The existing roof is a standing seam metal roof over 5" (R-27) rigid insulation with a roof pitch of 5" vertical to 12 horizontal. Standing seam metal system to have a LEED compliant reflective coating.

Vertical walls at existing roof dormers are metal panels with concealed fasteners.

Roof drainage is collected via guttering system and conveyed to the stormwater drainage infrastructure.

DOOR AND WINDOW FENESTRATION

Existing doors and windows will remain.

INTERIOR PARTITION SYSTEM

New Interior partitions, have been selected to meet structural, sound, fire and durability characteristics specific to a space or wall location within the footprint. Typical partition walls will be gypsum board and metal stud framing with acoustical insulation and concrete masonry unit partitions. Interior design shall comply with UFC 3-120-10. Reference the Interior Design narrative herein.

OPENINGS

Interior partition openings are a hollow metal frame type. Typical interior doors are flush wood and hollow metal to match the wall types encountered. Typical exterior doors are hollow metal at service entrances and heavy duty, wide stile aluminum at building entrances.

CEILING SYSTEMS

Ceilings will be suspended metal grid systems with mineral fiber lay in tiles to provide the most appropriate combination of acoustics, light reflectance and criteria for high recycled content. Head and shower rooms will be finished with moisture resistant, gypsum board ceilings. Mechanical, electrical and telecom rooms will have partitions that extend to the deck with unfinished, exposed structure above.

FLOOR SYSTEMS

Floor systems will consist of typical commercial finishes on concrete slabs.

NOISE AND ACOUSTIC CONTROL

Spaces will be evaluated, and appropriate STC rated wall and ceiling systems will be employed where directed.

SECURITY REQUIREMENTS

No special physical Security requirements have been identified.

ANTITERRORISM (AT) MITIGATION

- Antiterrorism (AT) Mitigation requirements will be continually evaluated throughout the design process and adjusted to meet the applicable requirements in the following UFCs.
- UFC 4-010-01, DoD Minimum Antiterrorism Standards for Buildings with Change 3 dated 25 May 2024
- UFC 4-020-01, DoD Security Engineering Facilities Planning Manual dated 11 September 2008
- The facility site is exempt from Standard 1: Standoff Distance as an existing building located within a controlled perimeter.
- Exterior glazing is to be laminated (UFC 4-010-01, Chapter 3, Standard 10)
- Exterior doors and frames are to be steel and out swinging (UFC 4-010-01, Chapter 3, Standard 12)
- Air intakes are to be a minimum of 10 Feet above the ground (UFC 4-010-01, Chapter 3, Standard 16)
- Emergency are distribution shutoff (UFC 4-010-01, Chapter 3, Standard 18)
- Equipment Bracing (UFC 4-010-01, Chapter 3, Standard 19)

- Mass notification (UFC 4-010-01, Chapter 3, Standard 21)

HAZARDOUS MATERIAL TESTING

ECS Carolinas has performed lead, asbestos, and mold testing at building TT113. During testing, no lead-containing paints asbestos containing materials, or evidence of mold were found. See Asbestos, Lead Paint, and Mold Assessment Survey for additional information.

HIGH PERFORMANCE AND SUSTAINABLE BUILDING REQUIREMENTS

High Performance and Sustainable Building (HPSB) Requirements will be continually evaluated and coordinated during design in accordance with UFC 1-200-02, High Performance and Sustainable Buildings Requirements and the Integrated Design Process guidelines of ASHRAE 189.1. Specific requirements for each building are as follows:

Building TT113 is greater than 10,000 GSF and greater than \$3M construction cost. Per Table 1-1, Compliance Requirements/Thresholds, this building is required to comply only with Chapters 1, 2, 4 and applicable appendices.

Reference UFC 1-200-02, High Performance and Sustainable Buildings

- Chapter 1 Requirements include:
 - Life Cycle Cost Analysis (LCCA) and cyber security compliance
- Chapter 2 Requirements include:
 - Commissioning in accordance with ASHRAE 189.1
 - Optimized energy performance to achieve at least 30% energy consumption reduction as compared to ASHRAE 90.1 baseline
 - Energy Efficient Products
 - On-Site Renewable Energy or LCCA Documentation
 - Solar Domestic hot water to meet 30% of annual hot water demand or LCCA Documentation
 - Utility metering
 - Indoor water use reduction and metering
 - Indoor air quality
 - Reduce environmental impact of materials
 - Storage and collection of recyclables
 - Construction waste diversion
- Chapter 4 Requirements include:
 - HPSB compliance checklists and reporting
 - Note: Third Party Certification is not required

It is not anticipated that on-site renewable energy or solar domestic hot water LCCA's will indicate implementation of these types of systems. The preferred approach is energy conservation through efficient design.

CIVIL BASIS OF DESIGN

The following Federal DoD criteria govern the civil design herein:

UFC 3-201-01	Civil Engineering, 20 December 2022
UFC 3-230-01	Water Storage, Distribution, and Transmission, with Change 3, 01 July 2021
UFC 3-240-01	Wastewater Collection and Treatment, with Change 4, 01 October 2024
UFC 3-250-04	Standard Practice for Concrete Pavement Construction, 29 January 2024
UFC 3-600-01	Fire Protection Engineering for Facilities, with Change 6, 06 May 2021
UFC 4-010-01	DOD Minimum Antiterrorism Standards for Buildings, with Change 3, 24 May 2024
ASTM	American Society for Testing and Materials
ADA/ABA	Americans with Disabilities Act (ADA) and Architectural Barrier Act (ABA) Accessibility Guidelines
NCDENR	NC Department of Environmental & Natural Resources

PROJECT SCOPE AND DISCUSSION

This project proposes the renovations of TT113 at Marine Corps Base, Camp Lejeune, North Carolina. The building is currently and will continue to be used as a Child Development Center.

The civil scope of work for this work included the following:

1. Completion of a physical and topographic survey of the project area.
2. Utility locate for the project area.
3. Development of plans associated with building renovations including the construction of a new mechanical building and mechanical yard.
4. Extension of utilities to serve the new mechanical building including:
 - a. Public Sewer
 - b. Public Water
 - c. Natural Gas
5. Designation of parking spaces for future installation of Electric Vehicle charging equipment.

EXISTING SITE CONDITION

Building TT113 houses one of the Child Development Centers at MCB, Camp Lejeune, North Carolina. The facility was originally constructed in approximately 2010. Public utilities are available in the area and serve the facility.

The site has a gentle topographic relief sloping away from the facility. Stormwater runoff is collected in storm drain system which discharges to a stormwater retention pond located on the western side of the site.

PROPOSED DEMOLITION

The proposed exterior demolition includes the removal of existing HVAC units and their associated electrical services.

PROPOSED CONSTRUCTION

The proposed site improvements for this project include the following major elements:

1. Construction of a new mechanical building and mechanical yard.
2. Installation of support utilities to serve the new mechanical building including:
 - a. Domestic Sewer Service
 - b. Domestic Water Service
 - c. Natural Gas Service
3. Designation of parking spaces for future installation of Electric Vehicle charging equipment. Pathways for the electric service to the charging equipment will be provided under this project.

UTILITY CONNECTIONS

Water Supply

The existing domestic water service to the facility will remain. The backflow preventer in the existing mechanical room will be replaced with a backflow preventer and enclosure installed exterior of the building.

A new 1" domestic water service will be provided to the new mechanical building.

The existing sprinkler service to the facility will remain.

Sanitary Sewer

A new 3-inch sewer service will be extended to serve the new mechanical building.

PROJECT PERMITTING

Storm Drainage System

This project is located within the boundary of an existing NCDEQ Stormwater Permit. The proposed construction will result in the creation of 1,530 square feet of additional permeable area.

A stormwater permit modification will be secured associated with the proposed BUA modifications.

Erosion Control

An Erosion and Sedimentation Control Permit is not required to properly execute this project since less than one acre of land disturbing activities will occur. Appropriate erosion control devices are included in the design to restrain erosion and sedimentation during construction.

STRUCTURAL BASIS OF DESIGN

REFERENCE DOCUMENTS

Note: The active governing codes are as prescribed in UFC 3-301-01 effective on the date of this BOD.

UFC 1-200-01	DoD Building Code, with Change 3, 26 February 2024
UFC 3-301-01	Structural Engineering, Change 2, 04 September 2024
UFC 4-010-01	DOD Minimum Antiterrorism Standards for Buildings, with Change 3, 24 May 2024
IBC 2021	International Building Code
ACI 318-19	Building Code Requirements for Structural Concrete
ASCE 7-16	Minimum Design Loads for Buildings and Other Structures
ACI 360R-10	Guide to Design of Slabs-on-Ground
AWS D1.1-20	Structural Welding Code

DESIGN LOAD CRITERIA

Risk Category II

LIVE LOADS:

- Roof = 20psf
- Catwalk = 40 psf
- Floor = 100 psf
- Mechanical Access Platform & Catwalk = 125 psf

SNOW LOAD:

Importance Factor = 1.0

Ground snow load, $P_g = 10$ psf

Snow Exposure Factor, $C_E = 1.0$

Thermal Factor, $C_t = 1.0$

WIND DESIGN DATA:

Ultimate Wind Speed = 144 mph

Wind Exposure Category = C

SEISMIC DESIGN DATA:

Seismic Importance Factor = 1.0

Mapped Spectral Response Accelerations:

$S_s = 11.5\%g$

$S_1 = 5.5\%g$

Site Class = D

Seismic Design Category = B

STRUCTURAL SYSTEM

TT113

The existing building consists of load-bearing concrete masonry unit (CMU) walls supporting light-gauge metal trusses. The renovation of the building mechanical system includes the installation of a new catwalk within the roof trusses that will be constructed with light-gauge metal stud joists which span between the CMU corridor walls and support rectangular bar grating. Access to the catwalk will be provided at three locations with new light-gage metal framed dormers within the existing hip roof. A new exterior steel framed mechanical access platform will also be provided at the west end of the building.

MECHANICAL BUILDING

VERTICAL LOAD RESISTING SYSTEM

The roof system will consist of a steel deck diaphragm atop prefabricated light-gauge metal stud trusses which span between load bearing reinforced concrete masonry unit walls. The foundation will consist of shallow concrete spread footings.

WALLS

The structural exterior walls will be composed of reinforced concrete masonry units.

LATERAL FORCE RESISTING SYSTEM

The lateral force resisting system will consist of a steel deck diaphragm that will distribute lateral loads to reinforced concrete masonry unit shear walls.

FIRE PROTECTION BASIS OF DESIGN

Reference the Fire Protection Design Analysis provided as a separate document with this design submittal.

PLUMBING BASIS OF DESIGN

APPLICABLE DESIGN CRITERIA

UFC 3-420-01 Plumbing Systems (UFC)

PLUMBING SYSTEMS

The plumbing systems within the facilities will consist of the following plumbing fixtures:

- Water Heater

317-C Pollock Street | New Bern, NC 28560 | T: 252-637-6373 | F: 252.637.3259

www.mbfarchitects.com

- Backflow Preventer

The existing electric water heater that serves the Classroom Areas is undersized and inoperable at the moment. It will be replaced with a properly sized gas water heater. The existing recirculation pump and thermostatic mixing valve installed in this area will remain. The existing reduced pressure zone backflow preventer will be removed and reinstalled exterior of the building near the Mechanical Room it is presently located. This will allow space for a ceiling-accessible ladder to be installed in the Mechanical Room.

HOT WATER CALCULATIONS – CLASSROOMS

Mixed Water Temperature Equation: $P = (T_m - T_c) / (T_h - T_c)$

T_h = Supply hot water temperature

T_c = Inlet cold water temperature

T_m = Desired mixed water temperature

P is the hot water multiplier, which is expressed as a percentage.

$$P = (120 - 65) / (140 - 65)$$

$$P = 55 / 75 = .73$$

Fixtures:

(28) Lavatories x 6 GPH = 168 GPH

(1) Service Sinks x 10 GPH = 10 GPH

(54) Sinks x 5 GPH = 270 GPH

Total - 448 GPH x .73 (Demand Factor) = 327.04 GPH

The water heater specified is 100 gallons storage with 250,000 BTU input and can recover a total of 323 GPH combined total at a 90-degree rise. With 80 gallons readily available + 323 GPH recovery, this heater can provide 403 GPH for the peak demand hour.

MATERIALS

Domestic water piping will be copper tubing, ASTM B88; Type L with solder joint fittings fabricated with code approved “lead free” solder for above ground piping and type K copper tubing for below grade applications.

Glass fiber insulation with all service jacket will be applied to all domestic cold and hot water piping.

Valves will be solder type, ball, for use with the copper tubing, Class 125 in accordance with MSS SP-80.

TESTING

Domestic water piping will be tested in sections at 100 psig for two hours.

Disinfection will be in accordance with AWWA C651 using a solution containing a minimum of 50 parts per million of available chlorine.

MECHANICAL BASIS OF DESIGN

PROJECT DESCRIPTION

This is a renovation project of an approximately 32,000 sq.ft. child development center located at Bldg. TT-113 Iwo Jima Blvd., Tarawa Terrace, Jacksonville, North Carolina.

DESIGN CONDITIONS

Outdoor Summer Design temperatures:	91.0°F DB / 77.0°F WB (1% ASHRAE)
Outdoor Winter Design temperature:	22.0°F DB (99.6% ASHRAE)
Indoor Design Conditions:	Occupied: 76°F Summer / 70°F Winter Unoccupied: 82°F Summer / 60°F Winter
Indoor Design Relative Humidity:	30% to 60%

DESIGN CRITERIA

The design shall be in accordance with the latest editions of, but not limited to, the following design criteria:

International Mechanical Code – 2024 Edition

NFPA 54	National Fuel Gas Code, 2024 edition
NFPA 90A	Standard for the Installation of Air-Conditioning and Ventilation System, 2024 edition
UFC 1-200-01	DoD Building Code, with Change 3, 26 February 2024
UFC 1-200-02	High Performance and Sustainable Building Requirements, with Change 2, 01 June 2022
FC 1-300-09N	Navy and Marine Corps Design Procedures, 17 May 2024
UFC 3-400-02	Design: Engineering Weather Data, with Change 2, 20 September 2024
UFC 3-401-01	Mechanical Engineering, with Change 1, 01 October 2015
UFC 3-410-01	Heating, Ventilating, and Air Conditioning Systems, with Change 9, 09 January 2024
UFC 4-010-01	DOD Minimum Antiterrorism Standards for Buildings, with Change 3, 24 May 2024
UFC 4-410-02	DDC for HVAC, Change 2, 12 April 2021
ASHRAE 15	Safety Standard for Refrigerated Systems, 2016 edition
ASHRAE 36	High-Performance Sequences of Operation for Ventilation Systems, 2021 edition
ASHRAE 55	Thermal Environmental Conditions for Human Occupancy, 2023 edition
ASHRAE 62.1	Ventilation for Acceptable Indoor Air Quality, 2022 edition
ASHRAE 90.1	Energy Standard for Sites and Buildings except Low Rise Residential Buildings, 2022 FC
SMACNA	HVAC Duct Construction Standards - Metal and Flexible, 4 th Edition
SMACNA	HVAC Air Duct Leakage Test Manual, 2nd Edition
Camp Lejeune Mechanical Design Guidance – Aug 2022	

DEMOLITION

Demolition work on this project includes the demolition and removal of the entire existing HVAC system serving the building. The existing main HVAC system is a Variable Refrigerant Flow (VRF) system comprised of above ceiling VRF fan coil units providing zone level DX cooling and heating with three (3) energy recovery ventilators (ERV) with energy wheel and DX coil for preconditioning of ventilation air ducted directly to fan coil return ductwork. The VRF DX heat pump condensing units are mounted on the exterior

of the building. All above ceiling associated refrigerant and condensate piping, branch controllers, ductwork, controls and wiring are to be demolished and removed. All local exhaust fans and electric unit heaters and associated ductwork, controls and wiring are to be demolished and removed.

MECHANICAL SYSTEM

The primary HVAC system for this building is 4-pipe chilled water / hot water system with a constant volume Dedicated Outdoor Air System (DOAS) unit and 4-pipe Variable Air Volume (VAV) units with single duct terminal boxes to serve the various zones. Two (2) air-cooled chillers will be in the Mechanical Yard and chilled water will be piped above ground through a buffer tank to the new Mechanical Building. Chilled water system will be pumped primary-secondary underground into the building and to the DOAS unit and all FCU air handlers in the new mezzanine above the main corridors. A new hot water system with natural gas-fired condensing boilers will be utilized, piped primary/secondary underground to the DOAS unit, VAV air handlers. Ventilation air will be pre-conditioned by the interior DOAS unit with a supply fan, reheat hot water coils, chilled water coils, pre-heat hot water coils, a fixed plate, total enthalpy heat exchanger and an exhaust fan. The ventilation air is delivered directly to the FCU air handler mixing section. The amount of ventilation air will be measured, to ensure sufficient quantities are brought into the building to meet ASHRAE 62.1 – 2019 requirements to maintain indoor air quality, and to positively pressurize the corridors and all rooms with exterior openings.

General

Air distribution will be via overhead galvanized sheet metal ductwork designed in accordance with SMACNA Duct Construction Standards. Duct will be wrapped with 2", 1-lb. exterior insulation, no duct liner shall be used. The ductwork systems are designed to provide a low noise environment. Air distribution shall be ceiling mounted. Duct construction shall be for low pressure on all units.

We have provided separate and independent ductless split-system DX cooling units for all Telecommunication and Electrical rooms.

All air systems shall have a full Test and Balance performed to verify performance match's design.

Exhaust shall be provided in all Toilet rooms, Shower Rooms and Janitors closets and shall be routed through the DOAS units and then out the building.

Dryers shall be individually ducted to a lint trap, with an opening through a louver to the exterior and an access panel for clean-out.

ATFP

All applicable requirements of UFC 04-010-01 will be followed. All outside air intakes shall be a minimum of 10' above grade. Low leakage dampers will be installed at all building exterior wall penetrations by the HVAC system. An emergency stop button will be located inside the facility to shut down all air moving systems.

BUILDING AUTOMATION SYSTEM

All control and monitoring points will be connected through a single complete building automation system (BAS) suitable for direct digital control (DDC) of all HVAC system components within the building. The

control system within the building shall utilize an open protocol implementation of BACnet technology and will utilize ANSI/CEA 709.1B as the communication protocol between DDC hardware devices. All utility meters shall employ the BACnet protocol, with the water and gas meters outputting to the building DDC system and the electric meters outputting directly to the basewide electric metering system. The DDC system will have the ability to limit the operating hours of any equipment, as well as perform setback functions when parts of the building are unoccupied. The DDC system will be compliant with all Cybersecurity requirements. The Mechanical Room shall be provided with a local display panel for monitoring and schedule program changes. The BAS shall be provided with a router to communicate all building points to the basewide EMCS located in Building 24.

SUSTAINABLE DESIGN

The building HVAC proposed system type was pre-determined utilizing the Life-Cycle Cost Analysis report prepared by Jacobs Engineering for Eastern North Carolina US Marine Corps Installations.

The HVAC systems for this project will be designed in accordance with UFC 1-200-02. An Energy Model will be used to show that the facility has achieved energy savings above a baseline system. The Energy Model has been run utilizing the Carrier HAP program, version 6.2. The selected system will be compared with the ASHRAE 90.1-2019 Appendix G Baseline system.

ELECTRICAL BASIS OF DESIGN

REFERENCE DOCUMENTS

Note: The active governing codes are as prescribed in FC 1-300-09N effective on the date of this task order.

UFC 3-501-01	Electrical Engineering, 27 November 2024
UFC 3-520-01	Interior Electrical Systems, 16 January 2025
UFC 3-530-01	Interior and Exterior Lighting Systems and Controls, Change 1, 15 December 2023
UFC 3-550-01	Exterior Electrical Power Distribution, Change 3, 01 Nov 2019
UFC 3-560-01	Operation and Maintenance: Electrical Safety, Change 3, 15 May 2023
UFC 3-575-01	Lightning and Static Electricity Protection Systems, Change 1, 01 Oct 2021
NFPA 70	National Electric Code, 2023 edition
NFPA 101	Life Safety Code, 2024 edition
NFPA 780	Standard for the Installation of Lightning Protection Systems, 2023 edition
Camp Lejeune Electrical Policies_v20220308	
Applicable UL standards	
Applicable NEMA standards	

EXTERIOR ELECTRICAL SYSTEMS

A new electrical system will connect the existing 12.47 kV underground primary owned by the power company. The existing electrical service will be reused to the greatest extent possible for the Child Development Center. The existing 300 kVA pad-mounted transformer is owned and provided by the power company. Contractor must coordinate with power company for increased loads included in this project. Secondary service conductors for the mechanical building shall be fed from the PMT to a

service-entrance rated, 480 volt, 3-phase, 400 amp main breaker panelboard inside the building. Spare conduits will be installed in the service duct banks to allow for future flexibility. A lightning protection system per NFPA 780 will be provided for this building since there was already one existing. New materials will be needed for the new roof, but existing downleads and grounding may be reused if contractor deems appropriate. Proper testing must be completed to confirm.

INTERIOR ELECTRICAL SYSTEMS

The electrical building distribution system shall be reused and adjusted for all new equipment connections. All feeders shall be sized for a maximum voltage drop of 2%, while branch circuits shall not exceed a voltage drop of 3%. All conductors shall be copper, and MC cabling is not allowed, except per UFC restrictions. All conductors shall be installed in conduit with a minimum conduit size of 3/4". All electrical devices in this facility shall be U.L. listed and labeled.

Interior lighting shall be designed in accordance with the Illuminating Engineering Society (IES) recommendations. Lighting levels shall meet UFC and user requirements. All new light fixtures shall be energy-efficient LED type, with 0-10 volt dimming capability. New lighting controls, including occupancy sensors and dimmer switches will be provided.

TELECOMMUNICATIONS BASIS OF DESIGN

APPLICABLE DESIGN CRITERIA

UFC 3-580-01 Telecommunications Building System Design, 21 November Jun 2024
Camp Lejeune Base Specification 27 10 00, v04/22
Camp Lejeune Base Specification 33 82 00, v08/22
Applicable EIA and TIA standards (latest editions)
Applicable UL standards (latest editions)
NFPA 70 National Electric Code, 2023 edition

EXTERIOR TELECOMMUNICATIONS SYSTEMS

Existing OSP infrastructure and cabling is existing and shall remain for the CDC building.

INTERIOR TELECOMMUNICATIONS SYSTEMS

The interior scope for this project consists of removing and replacing outlet faceplates since the wall finishes are being redone. All cabling must be retested at the completion of the project to ensure compliance. All telecommunications requirements shall adhere to the applicable design criteria and base standards.

INTERIORS BASIS OF DESIGN

INTERIOR DESIGN CRITERIA

Interior Design work will comply with the latest edition of the following codes and standards:

FC 1-300-09N	Navy and Marine Corps Design Procedures, 17 May 2024
UFC 1-200-01	DoD Building Code, with Change 3, 26 February 2024
UFC 1-200-02	High Performance and Sustainable Building Requirements, with Change 2, 01 June 2022
FC 1-300-09N	Navy and Marine Corps Design Procedures, 17 May 2024
UFC 3-120-01	Design: Sign Standards, with Change 4, 01 February 2023
UFC 3-120-10	Interior Design, with Change 2, 15 June 2021
UFC 3-190-06	Protective Coatings and Paints, 01 March 2021
UFC 4-740-06	Youth Centers, with Change 1, 21 March 2022

STRUCTURAL INTERIOR DESIGN

The TT113 Child Development Center Renovation at Camp Lejeune, NC requires interior finishes that have a balance between aesthetics and sustainability, as well as durability that can withstand the inherent day-to-day operating environment for staff and patrons utilizing the facility. To meet these objectives, the SID package will use all the general requirements for interior finishes that have been applied in similar facilities, as well as best practices recommended by manufacturers for areas requiring enhanced durability. SID finishes will be vetted and compared to any specific items or requirements requested by the activity.

The SID package will provide a variety of floor finishes throughout including attached cushion carpet tile and luxury vinyl tile. Interior partitions will be painted drywall.

Acoustical ceiling tiles on a suspended grid will provide high quality sound absorption in designated areas with easy access to mechanical and plumbing above. The square edge lay-in tiles are a moisture resistant mineral fiber and have a minimum NRC of 0.85, CAC of 35, and a light reflectance of 0.86. Restrooms will have painted gypsum ceilings.